Comparative Study of Lightweight CNN Models on Fashion-MNIST

GitHub repository:
https://github.com/2766089255-dot/Comparative-Study-of-Lightweight-CNN-Models-on-Fashion-MNIST

Abstract

This experiment uses three mainstream lightweight convolutional neural networks—MobileNetV2, ShuffleNetV2, and EfficientNet-B0—to compare results on the Fashion-MNIST dataset. All models were trained with the same parameters: 30 training epochs, a stochastic gradient descent optimizer, and a cosine annealing learning rate. The experiment compares different models based on accuracy, parameter count, floating-point computation cost, and inference time. Results show that all three networks achieve a similar accuracy of approximately 93%. Since ShuffleNetV2 performed best in the comparison, it is considered an ideal choice for resource-constrained devices. Although EfficientNet-B0 has 4.14 million parameters and 29.78 million floating-point operations, its accuracy was not improved, indicating that this network may be redundant for this simple dataset. This study provides a practical reference for the selection of lightweight models in mobile vision tasks.

1.Introduction

The proliferation of mobile devices and embedded systems has fueled market demand for accurate and computationally efficient image classification models. Lightweight convolutional neural networks such as MobileNet5[1], ShuffleNet[2], and EfficientNet[3] have emerged to achieve this balance. However, most performance comparisons in existing literature are based on large benchmarks like ImageNet, which may not be directly applicable to smaller, domain-specific datasets.

To fill this gap, we compared three representative lightweight architectures on the Fashion-MNIST dataset [4]. This dataset contains 70,000 28×28 pixel grayscale images of clothing, divided into 10 categories, and is generally considered a more challenging alternative to the original MNIST.

The core contributions of this study are threefold. First, by maintaining complete consistency in all training conditions, we ensured the uniformity and repeatability of the comparison results. Second, we evaluated the models using multiple metrics, including accuracy, number of parameters, number of FLOPS, and inference time. Finally, we analyzed the trade-off between model complexity and actual performance, providing a basis for model selection in practical applications.

The structure of this paper is as follows:

Section 2 provides a brief overview of relevant research.

Section 3 details the experimental methods, including model selection, training configuration, and evaluation metrics.

Section 4 presents and discusses the experimental results; and Section 5 summarizes the entire paper and proposes future research directions.

2.Related Work

2.1 Lightweight CNN Architectures

Lightweight convolutional neural networks started with MobileNetV1[5], which reduced computation by introducing depth-wise separable convolutions. Then, MobileNetV2[1] introduced inverse residuals and linear bottleneck structures to improve computational efficiency. Meanwhile, ShuffleNetV1[2] introduced group convolutions and channel shuffling techniques, and ShuffleNetV2[6] optimized the network design through four practical rules. Efficient-Net[3] used different methods to achieve high accuracy with a small number of parameters by adjusting the width, depth and resolution of the neural architecture.

2.2 Fashion-MNIST Dataset

Fashion-MNIST[4] was proposed as an alternative to MNIST, which can contain more realistic and complex patterns. Today, it has become the standard for evaluating the performance of image classifiers on small-scale data.

2.3 Existing Comparative Studies

Several studies have compared lightweight models, but most have focused on Image-Net[7] or CIFAR[8]. There are few studies that provide a systematic comparison of Fashion-MNIST under strict control. This study aims to provide a comparison for such cases.

3. Methodology

3.1 Model Selection

This experiment selected three widely-used lightweight models: MobileNetV2 with a width multiplier of 0.5, ShuffleNetV2 with a scale factor of 1.0, and EfficientNetB0. Since EfficientNetB0 is designed for large-sized images such as Image-Net, this study adjusted the backbone convolution stride from 2 to 1 to prevent the loss of texture or edge information during sampling, which would affect the results.

3.2 Training Setup

All models were trained on the Fashion-MNIST training set and tested using the official test set. Images were resized to 28×28. To match the expected input format of the models, Grayscale was used to convert to a three-channel format. Data augmentation included random horizontal flipping and random cropping. Pixel values were normalized using the mean and standard deviation.

All models used the same training configuration: an SGD optimizer with momentum of 0.9 and weight decay of 5×10^{-4}; an initial learning rate of 0.1, decaying after 5 epochs using a cosine annealing scheduler; a batch size of 128; a training period of 30 epochs; and a cross-entropy loss function. To obtain reliable statistical results, each model was trained three times using a different random seed.

3.3 Evaluation Metrics

This experiment uses four metrics to evaluate model performance: the average and standard deviation of three runs, the total number of trainable parameters, the FLOPs of a single forward propagation with an input size of $3\times 28\times 28$, and the average inference time per image, measured over 200 iterations after 10 warm-up runs.

4. Results and Analysis

4.1 Quantitative Results

The average performance of the three models in three independent runs is shown in Table 4.1.

Table 4.1: Performance comparison of lightweight models on Fashion-MNIST

Model	Accuracy (%)	Parameters	FLOPs	Inference Time (ms)
MobileNetV2	93.12 ± 0.05	815,780	8.092M	8.96 ± 0.66
ShuffleNetV2	93.13 ± 0.07	1,356,104	11.548M	9.27 ± 0.33
EfficientNet-B0	93.13 ± 0.09	4,135,648	29.781M	13.95 ± 0.57

4.2 Accuracy Analysis

The results show that the top-1 accuracy of the three neural networks is similar, all reaching around 93.1%, and the standard deviation is less than 0.1%. This indicates that these lightweight architectures have considerable learning capabilities on the Fashion-MNIST dataset, and that this dataset is relatively simple for this type of convolutional neural network. These subtle differences are entirely within a manageable range.

4.3 Efficiency Analysis

As shown in Figures 4.1, 4.2 and 4.3, MobileNetV2 has the fewest parameters and floating-point operations, and the second fastest inference time, achieving the best balance between accuracy and efficiency.

ShuffleNetV2 has slightly more parameters and floating-point operations, but its inference speed remains fast. Its accuracy is the same as MobileNetV2.

In contrast, EfficientNet-B0 has the largest number of parameters and floating-point operations, but the slowest inference speed. Despite its higher complexity, it does not improve accuracy, indicating that this model wastes resources on inference with small datasets.

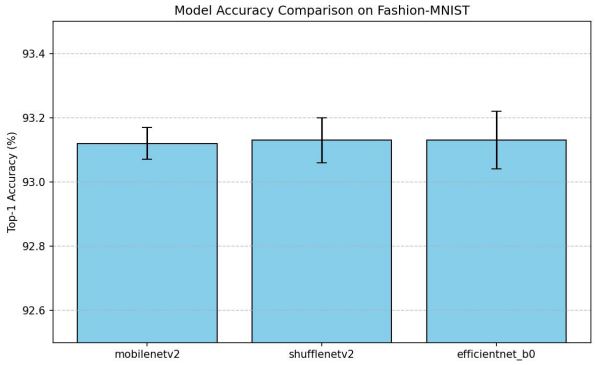

Figure 4.1 Accuracy Comparison Bar Chart

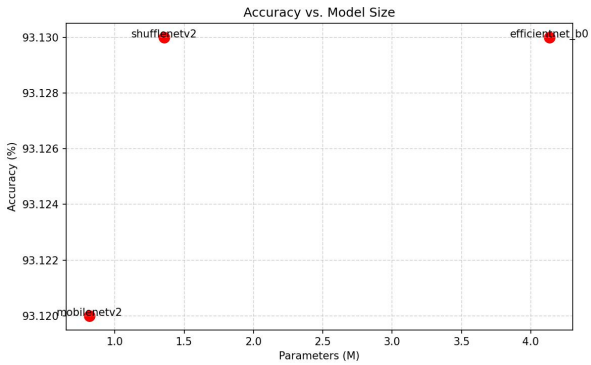

Figure 4.2 Relationship between accuracy and number of parameters

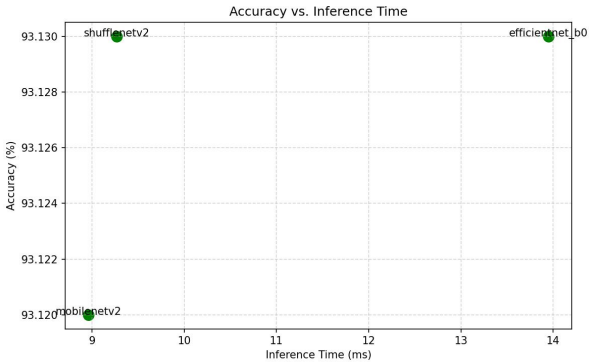

Figure 4.3 Relationship between accuracy and reasoning time

5. Conclusion and Future Work

This experiment systematically compared three lightweight convolutional neural network models on the Fashion-MNIST dataset. Under identical experimental conditions, the results showed that all models performed similarly in accuracy, but their efficiency differed significantly. MobileNetV2 achieved the best balance between performance and resource consumption, while EfficientNetB0 proved too complex for this task and unsuitable for small datasets. ShuffleNetV2 also performed well, but its training cost was higher compared to MobileNetV2.

Future research could extend this comparison to other small-scale datasets and incorporate state-of-the-art architectures such as MobileNetV3 or EfficientNetV2. Furthermore, deploying these models on real mobile hardware and measuring energy consumption would provide more practical applications. Model compression techniques could also be further explored to improve their efficiency.

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